

Thought-Density Hypertime and the Extended Brain

A Falsifiable Architecture, Executable Preparticipant Protocol, and Adverse Synthetic Design Audit

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

An external cognitive system can store information, recommend actions, or control a device without becoming part of a person's continuing cognitive process. This paper defines a stricter candidate architecture. An **Extended Brain** is a reciprocal, adaptive human-machine system in which inspectable human, artificial, and shared-workspace states repeatedly change one another during a task. The associated **thought-density hypertime hypothesis** proposes that such coupling could increase useful cognitive updates that a person accesses, uses, and later recovers per unit of physical time. It does not propose that world-clock time or biological lifespan changes.

The historical program is separated from the scientific test. Public 2012 Neo Mind Cycle recordings establish the earlier feedback-loop, predictive-self, and computer-mediated neurofeedback ancestry. Private contemporaneous 2012 records separately preserve proposed brain-computer-interface, extended-brain or artificial-cortex, and sentient-artificial-brain branches; they document author-side genealogy but are not public 2012 priority proof. Public 2017 Neural Lace sources develop a distributed path from artificial neocortex to expanded thought workspace, higher cognitive throughput, concurrent thought, and altered time experience. A dated October 2021 audio record restates the closed sensor-computer-sensor loop and distinguishes NerveGear, External Cortex, and autonomous artificial-brain architectures; November records consolidate the three-project recollection while separately identifying quantum-gravity work as a new fourth branch. Public 2022 Git records fix the clearest compact hypertime statement. This genealogy establishes development history, not scientific validity.

Draft 2 froze five within-participant conditions: unaided baseline, ordinary tool, adaptive but nonreciprocal AI, reciprocal closed loop, and content- and latency-matched yoked replay. A human-integrated event must be novel, accessible, consequential, nonduplicative, timestamped, and provenance-labeled. Support requires three sequential gates: the lower confidence bound for reciprocal performance must exceed each strong control by 0.20 standard deviations; the lower bound for strict integrated-event rate must exceed each by 15 percent; calibration and agency must then remain within noninferiority margins. Time outcomes cannot rescue gate failure.

Draft 3 adds a coupled-state model, task and latency contracts, agency, memory-provenance, identity-continuity and decoupling measures, Monte Carlo power sensitivity, and synthetic parameter recovery. The power analysis exposed a design defect. Under the frozen stringent confidence-bound rule, 270 completers yielded only 46.5 percent sequential success in 2,000 simulations even when the true reciprocal advantage was 0.35 standard deviations and the event-rate ratio was 1.30. A sample-size sweep first exceeded 80 percent at 540 completers under those assumptions. This is not a universal required sample size: pilot variance, attrition, coder uncertainty, task correlation, and observed event dispersion must inform the final lock. It does show that 270 cannot remain the unexamined confirmatory target.

Draft 4 adds a source-bound empirical bridge and a stricter subjective-time preregistration. Human fMRI evidence shows that trial-by-trial duration-report biases during naturalistic viewing can be reconstructed from salient-event dynamics in stimulus-relevant visual cortex [13]. This supports measuring event structure and subjective duration together. It does not show that more thoughts per second universally slow felt time, that machine events become human thoughts, or that reciprocal coupling extends subjective lifespan. The study therefore retains performance and strict human-integrated event density as the sequential mechanistic gates and tests prospective duration, retrospective duration, felt passage, temporal resolution, and remembered boundaries as separate secondary outcomes.

A separate correctly specified linear simulation recovered reciprocal human-to-artificial and artificial-to-human cross-lag parameters across five latency levels with maximum absolute error 0.0201. At 50 ms, reciprocal estimates were 0.2568 and 0.2785, whereas yoked replay recovered human-to-artificial coupling of 0.0028. Recovered reciprocal coupling declined monotonically as latency increased, and every local linearization remained stable. These are synthetic identifiability results, not evidence that an Extended Brain exists. A second robustness audit shows why the confirmatory analysis must address attrition and misspecification: observed-risk weighting corrected a generated missing-at-random bias, but it could not correct outcome-dependent missingness; omitting shared-workspace state created false reciprocal paths in a nonreciprocal fixture; lag mismatch and measurement error attenuated generated coupling. No participant data have been collected or opened. The current contribution is a reproducible methods correction and failure-detection framework.

Draft 5 implements the pre-participant runner contract. It freezes 300 unique, hashed task instances across five conditions: 12 dynamic-planning, 8 evidence-integration, and 40 spatial-transformation instances per condition. Planning instances contain three deterministic hidden changes; evidence instances contain two planted conflict pairs; one quarter of spatial instances contain state-reconstruction probes with machine-checkable answers. A separate schedule contains 180 A/R/Y latency assignments with identical bin distributions and maximum standardized mean difference 0.0. The event-coder template exposes the five strict inclusion criteria and four provenance labels while withholding condition. Eleven validation checks pass. These are software-mechanics results only; no participant responses or outcomes were generated.

Scope and evidence boundary

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1. The problem

Computers already extend what people can retrieve and calculate. That fact does not settle where a cognitive process ends. A notebook can preserve a fact without tracking the changing state of its reader. A language model can answer a question without becoming part of the user's continuing agency. A decoder can translate brain activity into cursor motion while remaining one-way. The stronger Extended Brain proposal begins only when both sides of the system change through the interaction and when those changes matter to later perception, inference, memory, or action.

The motivating intuition can be stated without specialized language. Suppose a person and an artificial system solve a problem together. The person supplies goals, context, judgment, and embodied consequences. The artificial system supplies search, simulation, memory, or alternative representations. Each observes some consequences of the other's state and changes its next operation. If that reciprocal cycle becomes fast, stable, interpretable, and learned, the combined system may complete more useful cognitive transitions during the same minute than the unaided person.

The phrase **thought-density hypertime** names this proposed increase in integrated cognitive activity per physical interval. It does not mean that physical time slows. It does not mean that every generated token, neural spike, or software operation is a thought. It does not guarantee that the user feels time passing slowly. It does not imply biological immortality. Those distinctions are required because time experience is not a single variable.

2. Historical source genealogy

2.1 Public 2011-2012 ancestry and transcript authority

The earliest source layer explains why the later Extended Brain proposal was conceived as a loop rather than as a larger database. An owner-metadata-dated 2011 dialogue treats the self as a predictive pattern in the neocortex and then generalizes toward a whole-network predictive world model. Because the currently exposed YouTube date is 2025 and the supporting 2011 Studio export is not yet preserved in the public release, this is an authorship-genealogy anchor, not strong adversarial public-priority evidence.

The public July-August 2012 Neo Mind Cycle recordings provide the firmer operational ancestry. They describe EEG-derived state being transformed into light or sound, perceived by the same participant, and measured again; they also connect prediction, self-modeling, and an internally constructed world to that recursive feedback program. The later Extended Brain proposal enlarges this cycle from feedback about a person's measured state to a reciprocal machine workspace that can alter later inference, memory, and action.

Transcript versions are not interchangeable evidence. The original recording and its native platform timestamp or custody record are the quotation and provenance authorities. A clearer 2026 retranscription is a semantic reading aid whose load-bearing wording must still be checked against the recording. Older captions and transcripts are discovery and version-history aids only. Accordingly, this paper paraphrases unverified audio and uses quotation marks only for text-native records or recording passages with adequate readback.

2.2 A distributed private 2012 program

The three-branch engineering program is distributed across private, owner-controlled text records whose export metadata preserves 2012 creation dates. These records can establish continuity inside the author's custody chain, but they are not described as immutable public 2012 publications.

On July 15, 2012, a note proposes building a “perfect brain to computer interface” and linking two people so that they could learn each other's minds in a new way. This is an early version of the bidirectional interface problem: how can information produced by one mind become usable input to another without being reduced to an ordinary message?

On October 4, 2012, a preserved note states a goal of building an architecturally self-aware, compassionate AGI being. An August 17 note about machine-based brains with “real consciousness” independently corroborates that target. The language is speculative, but the research program is clear: the third branch was not merely better automation. It was an artificial brain proposed as a subject of experience.

On October 24, 2012, a preserved note proposes a cyborg or AI attachment that could create a brain extension, “super brain,” or “extended brain” 80 times larger in virtual space, making life “80 times more precise.” The factor is an imaginative illustration, not an empirical estimate, but the architecture and subjective-life consequence are explicit. A November 6 note describing a wearable AGI-Extenda-Brain provides further contemporaneous corroboration.

These three records do not contain the later finished causal chain in one passage. They do, however, establish that the BCI, external-cortex, and artificial-brain branches were all present in the 2012 Neo Mind Cycle period.

2.3 Program continuity, operation, and the external-cortex bridge

Two additional 2012 records clarify what Neo Mind Cycle meant in practice. A July 30 note explains the name directly: “Neo Mind” refers to neocortex, while “Cycle” refers to a neurofeedback feedback

loop. An October 21 note documents an operating neurofeedback program, neomindcycle.com, planned neurophysics teaching, and explicit interest in MEG, HEG, EEG, fMRI, DTI, TMS, and tDCS. These records support a real autodidactic and service program rather than a retrospectively invented label. They do not validate the note's intervention, drug, disease, or outcome claims, which remain outside this paper and require modern scientific review.

The external-cortex mission becomes more explicit before the compact 2022 hypertime passage. A June 2021 note identifies Neo Mind Cycle as the parent mission for SAN software architecture and the NerveGear BCI ideal. An October 23 note describes biosensing AR and BCI systems as a possible route for extending a person with a sentient external cortex. This is a design proposal, not evidence that available sensors have the required bandwidth or that two minds can literally merge.

March 2022 auto-transcribed notes supply two further conceptual bridges. One links parallel neural activity to effective response or frame throughput. The other asks whether an Extended Brain would require cross-system data conversion, temporal synchronization, a larger connected circuit, and a rule for preserving separate identities. Their quantitative and neuroscientific assertions are not treated as established results, and the recordings must be checked before any direct quotation. Their legitimate role here is narrower: they document why the confirmatory design below separates useful human- integrated event density, reciprocal contingency, identity, stability, and decoupling instead of using raw machine throughput as the outcome.

2.4 The public 2017 mechanism

The 2017 Neural Lace program adds the mechanism that turns external cognition into a subjective-time hypothesis.

A March 2017 article describes artificial neocortex as an extension of the brain's thought workspace. Neural Lace Podcast material places artificial neocortex after the interface and before autonomous artificial brains. A May 2017 discussion proposes switching among brain programs and achieving faster forms of calculation. A companion article describes thousands of thoughts appearing simultaneous while time seemed to slow. A June 2017 article discusses large concurrent cognitive traffic, variable cognitive timescales, and experiences resembling entire lifetimes compressed into minutes.

The complete architecture is distributed across these sources:

bidirectional neural interface

- > artificial neocortex or external cognitive workspace
- > increased concurrency or useful cognitive throughput
- > more structured cognitive events per physical interval
- > altered judgments or experience of time
- > potentially more experienced cognitive content within a life

The distributed form matters. Historical evaluation should not demand that a later compact sentence appear verbatim in every earlier source. It should also not pretend that any one 2017 source contains every arrow.

2.5 The October-November 2021 audio and text consolidation

An original-audio-bound October 11, 2021 Micah monologue supplies a mechanistic bridge without requiring the older transcript wording to be published. In bounded paraphrase, it describes measured brain state being transformed by a computer into sensory feedback, perceived by the participant, and measured again. It distinguishes three related but nonidentical architectures: NerveGear as an experiential read / write interface, External Cortex as augmentation, and an autonomous artificial brain. Its useful claim is reciprocal human-machine-cycle ancestry, not consumer-EEG semantic decoding or clinical efficacy.

A private text record created on November 30, 2021 explicitly lists:

1. Neural Lace or NerveGear, described as the ultimate BCI;
2. artificial cortex that expands the human mind into the computer; and
3. sentient artificial brains.

It says that the author dedicated his life to solving these three problems and describes a book written nine years later as a step toward them. A separate November 2021 record says that these life goals were realized in 2012.

This is strong author-side evidence about the intended unity of the program, but its date must be represented correctly. It is a 2021 joined recollection of a 2012 dedication, supported by separate 2012 component records. It is not a single contemporaneous 2012 declaration containing the exact later wording.

2.6 The compact 2022 formulation

The repository version of a0087z.md, fixed in June 2022, states the whole proposal compactly: an artificial cortex increases calculations and thoughts per second; more thoughts per second slow the experience of spacetime; greater awareness of each moment makes a lifespan feel longer.

That passage is the clearest single-source explanation. It is not the first appearance of the artificial-cortex or subjective-time mechanism. The 2012 component records and the public 2017 distributed architecture remain part of the genealogy.

2.7 The objectively dated fourth physics branch

The original program had three engineering and consciousness targets: NerveGear, Extended Brain, and sentient artificial brains. The later move from neural oscillation, duration, frequency, and inverse relationships into Quantum Gradient Time Crystal Dilation was not one of those original targets.

The surviving source record dates that separation. A November 11, 2021 note records the author's own "equation for quantum gravity." On November 13, the physics material is explicitly called a "new 4th section" of the book. A November 20 note repeatedly calls it a new equation and new theory of quantum gravity connected to neuroscience. Later 2022 records develop and name the Quantum Gradient Time Crystal Dilation formulation.

This does not claim that quantum gravity was absent from the author's earlier reading environment. It shows that the currently recovered authored theory appears as new in November 2021 and was structurally distinguished from the three original projects.

This distinction matters for SAN Book 2. The book can explain how the neuroscience story motivated a physics question without rewriting the 2012 vow as a quantum-gravity project.

3. Five architectures that must not be collapsed

3.1 Ordinary cognitive tool

A tool supplies information or transformation when asked. It need not model the user, adapt to the user, receive biological signals, or alter its future behavior through the shared task.

3.2 One-way neural decoder

A decoder maps biological signals into an output. The output may control a cursor, prosthesis, or software state. This is a BCI, but it is not yet a reciprocal external cortex.

3.3 Bidirectional neural interface

A bidirectional interface both reads from and writes information back to the user through sensory, electrical, or other channels. The feedback can support learning, but bidirectionality alone does not prove cognitive integration.

3.4 Reciprocal external cortex

An external cortex is a learned, reciprocal system in which:

- human state changes artificial processing;
- artificial state changes later human processing;
- the loop contributes to a continuing task;
- both sides adapt across repeated interactions;
- the system's contribution cannot be reproduced by a noncontingent replay with matched content and timing; and
- the user can inspect, interrupt, and leave the coupling.

This is the main scientific target of the paper.

3.5 Autonomous artificial brain

An autonomous artificial brain has its own continuing state, goals, learning, and possible welfare independent of a particular human coupling session. It must not be treated as a detachable organ merely because it can participate in an external-cortex loop.

Two autonomous minds may temporarily form a shared task cycle. That does not establish that they become one permanent person.

4. Prior-art boundary and distinctive join

The Extended Brain proposal sits at the intersection of older research areas. Its contribution cannot be stated responsibly by treating any one of those areas as new.

Research line	What it already contributes	What it does not by itself establish here
Man-computer symbiosis [1]	Cooperative human-machine intellectual work and complementary capabilities	A measurable human-integrated event rate, a subjective-time vector, or a contingency-matched test
Extended cognition [2]	Conditions under which reliably used external resources may participate in cognition	A closed-loop neural or AI implementation, a two-learner stability model, or empirical hypertime
Bidirectional BCI and artificial sensation [5,6]	Decoding, feedback, adaptation, learned sensory mappings, and embodied consequences	That an artificial system becomes an external cortex or increases experienced cognitive content
Adaptive automation and augmented cognition [10]	Physiological state can alter automation, and yoked controls can separate contingency from mere task switching	Reciprocal shared inference, human-integrated event adjudication, or thought-density hypertime

Research line	What it already contributes	What it does not by itself establish here
Co-adaptive interfaces [9]	Human and decoder can behave as two interacting learners whose adaptation requires stability analysis	Identity merger, a general cognitive workspace extension, or subjective-life expansion
Attention and time psychophysics [3,4]	Attention and arousal can alter duration judgments without necessarily improving temporal resolution	That more useful cognitive events make passage feel slower
Experiential time and event memory [7,8,11,12]	Duration, passage speed, event boundaries, and memory are related but separable	A single scalar internal time variable or a reciprocal-coupling mechanism
Salient-event reconstruction of subjective duration [13]	Trial-by-trial duration-report biases can be reconstructed from stimulus-relevant visual-cortex event dynamics during naturalistic viewing	That event density is identical to thought density, that higher event rates always slow passage, or that reciprocal human-AI coupling causes the effect

The proposed contribution is the **join**, not any component alone:

reciprocal, inspectable two-learner coupling

- + strict human-integration event rule
- + strong tool, adaptive-AI, and yoked-replay controls
- + separate performance, memory, time, agency, and safety outcomes
- + reversible identity and provenance boundaries

That join remains a hypothesis until it survives the pilot-gated, preregistered study. The term **external cortex** is architectural: it describes a candidate reciprocal functional role. It does not assert cortical tissue equivalence, consciousness transfer, or a new person. Likewise, **hypertime** names denser useful cognitive content within the same clock interval. It is not a physical time-dilation claim.

4.1 Prior-art nonclaims

This paper does not claim priority over symbiosis, extended cognition, adaptive automation, neural prostheses, co-adaptation, event segmentation, time perception, or artificial consciousness as broad fields. Its historical claim is limited to the recovered Blumberg genealogy. Its scientific claim is limited to a preregistrable architecture and discriminating experiment.

4.2 Why reciprocal contingency matters

An ordinary tool can improve performance. A responsive model can generate more information. An adaptive interface can personalize output. None of these facts demonstrates an Extended Brain. The distinctive causal question is whether the current person's changing state alters the artificial system and whether the altered artificial state then changes the person's next useful state. The yoked replay preserves output amount, class, and timing while breaking that contingency. If reciprocal performance and event-density gains survive the replay, they cannot be reduced to merely receiving more content or faster responses.

5. Operational definitions and event adjudication

Let physical or world-clock time be t . The study never infers that physical time changes. It asks how much task-relevant cognitive structure a person integrates during a fixed interval.

5.1 Candidate events

A candidate event is first detected from an objective record: a task-state transition; a new hypothesis or plan revision in the shared workspace; a human action or report that changes the task policy; an artificial proposal shown to the participant; or an explicit correction, rejection, or confidence update. Candidate generation is intentionally permissive. Counting is strict.

For candidate e , define five preregistered binary indicators:

- N_e : novelty relative to information immediately available before the event;
- A_e : human access, demonstrated by acknowledgment, use, rejection, or accurate later report;
- U_e : downstream use in a later prediction, memory, plan, or action;
- Q_e : adequate timestamp and origin provenance; and
- D_e : duplication or trivial semantic restatement under the frozen rules.

The strict human-integration indicator is

$$I_e = N_e A_e U_e Q_e (1 - D_e). \quad (1)$$

Every factor is required. Output displayed but ignored fails A_e or U_e . A repeated paraphrase fails $1 - D_e$. A hidden model state fails A_e and Q_e . Machine tokens and neural spikes are never counted by quantity alone. An event may originate with the human, artificial system, environment, or shared loop, but it enters the primary count only through the same human-integration rule.

For active task time Δt , strict event density is

$$D_H = \frac{\sum_{e=1}^K I_e}{\Delta t}, \quad (2)$$

and the reciprocal-to-control rate ratio is

$$G_H(R, c) = \frac{D_H^{(R)}}{D_H^{(c)}}, \quad c \in \{T, A, Y\}. \quad (3)$$

The primary rate is calculated per active minute, excluding frozen pause, technical-failure, and instruction intervals. Event origin is retained in the ledger rather than pooled away.

5.2 Adjudication

Two coders, blind to condition, independently score the strict event ledger. They train only on the excluded pilot set. The confirmatory ledger is accepted when Cohen’s kappa is at least 0.70. Below that threshold, coders retrain on pilot material, repeat the blind coding, and send remaining disagreements to a third blind adjudicator. Condition labels remain locked until the ledger and disagreements are frozen.

The strict detector is the only detector allowed to pass the mechanistic gate. Two sensitivity detectors are also reported: task-state-only events, which avoid semantic judgments but may miss cognitive integration; and boundary-plus-later-memory events, which require later event recall but may favor memorable over useful transitions. Sensitivity detectors may challenge the interpretation. They cannot rescue a failed strict analysis.

5.3 Outcome vector

The study keeps the outcome vector explicit:

$$\mathbf{Y} = (P, D_H, D_{\text{pro}}, D_{\text{retro}}, V_{\text{pass}}, R_T, M_E, C, A, F), \quad (4)$$

where P is task performance, D_H strict integrated-event density, D_{pro} prospective duration, D_{retro} retrospective duration, V_{pass} felt passage speed, R_T temporal resolution, M_E remembered event-boundary density, C confidence calibration, A agency, and F fatigue or effort.

No scalar “subjective time” replaces this vector. Biological lifespan is not an outcome. Increased performance does not imply altered time experience. Retrospective duration does not imply finer temporal resolution. More remembered boundaries do not establish more online thought. Felt passage may accelerate while retrospective duration expands.

6. What hypertime could mean

The strongest defensible interpretation is not “time slows.” It is:

During the same physical interval, a reciprocal cognitive system may enable a person to integrate, use, and later recover more structured cognitive events.

Several experiences could follow, and they need not move in the same direction.

First, an interval may be judged longer retrospectively because it contains more remembered events or boundaries. Second, a person may judge an attended stimulus as lasting longer prospectively. Third, the passage of time may feel faster when events change rapidly. Fourth, temporal discrimination may remain unchanged even when a later memory makes the interval seem expanded. Fifth, performance may improve while all subjective-time measures remain unchanged.

Existing psychophysics makes this separation mandatory. Frightening events can produce retrospective duration expansion without improved temporal resolution. Naturalistic videos with more events can be judged longer while their passage feels faster. Event boundaries can influence retrospective duration through memory. The original phrase “more thoughts per second makes time feel slower” is therefore a directional hypothesis, not a definition.

Sherman et al. provide a particularly close measurement bridge without closing the mechanism [13]. Their salient-event model reconstructed trial-by-trial biases in duration reports from visual-cortex activity while control sensory regions did not predict those biases. The measured construct was perceptual event change during video viewing, not thought count, reciprocal coupling, or subjective lifespan. Draft 4 therefore treats salient-event accumulation as a rival-compatible empirical precedent: it justifies an event-sensitive analysis and also requires the Extended Brain study to show that any time effect depends on strictly human-integrated events rather than stimulus change, output volume, novelty, arousal, or memory alone.

7. Reciprocal coupling as a causal boundary

Let the task-relevant human state, artificial-system state, inspectable shared workspace, and communication channel change together over repeated exchanges. Calling that arrangement an Extended Brain requires more than fast output or good assistance. The human’s current state must alter the artificial system’s later useful state, and that altered artificial state must in turn alter the human’s later useful state after common inputs are controlled.

The experiment therefore treats bidirectional, current-person contingency as a causal boundary rather than assuming it from correlation. Condition A can be highly capable without passive human-state coupling. Condition Y preserves the amount, class, and timing of an earlier reciprocal session while breaking contingency on the current participant. Those controls distinguish reciprocal coupling from receiving more content, faster responses, personalization, or a convincing replay.

Stability is part of the boundary. A fast positive feedback loop can amplify error, confidence, emotional arousal, fatigue, or dependence. A useful system must therefore combine bounded latency, calibrated uncertainty, reversible control, inspectable provenance, and successful decoupling. The numbered estimands below operationalize these requirements; this section introduces no additional metric that can be selected after observing results.

8. Executable preparticipant protocol and pilot-gated confirmatory design

8.1 Design and sample

The protocol is a randomized, within-participant, five-condition crossover. A 24-completer pilot is used only to debug tasks, estimate nuisance variance and coder reliability, select the declared lag window, train event coders, and exercise the analysis implementation. Pilot outcomes are never pooled into confirmation. The earlier 270-completer target is retained as a version-history design point, not as the final sample-size decision. Under the declared strong synthetic scenario it is underpowered; the first swept design above 80 percent was 540 completers. The final confirmatory sample and MNAR sensitivity range must therefore be computed from pilot-derived estimates and frozen before confirmatory enrollment. Incomplete sessions are handled only under the preregistered rule, with attrition reported by condition and reason.

Each condition occurs on a separate session at least 24 hours apart. A balanced Williams design distributes first-order carryover, while task-family order is counterbalanced independently. Participants complete criterion-based practice before randomization; practice trials are excluded. Before the pilot, the study architecture, task families, candidate estimands, event rules, exclusions, and diagnostic suite are version-locked. Before the first confirmatory participant, the final task instances, system prompts, model versions, latency envelopes, output budgets, sample size, nuisance estimates, MNAR range, scoring, and analysis code are hash-locked.

The participant is the independent unit. Participants cannot be blind to the presence of an aid, but they are not told whether a responsive-looking session is reciprocal or yoked. Outcome scorers and event coders remain blind until their ledgers are frozen.

8.2 Conditions

1. **B, unaided baseline:** task interface and scratch workspace without an external cognitive aid.
2. **T, ordinary tool:** static reference, calculator, and search over a frozen task corpus, with no user model or adaptation.
3. **A, adaptive nonreciprocal AI:** the same model family and output budget as the reciprocal system, but only explicit prompts and intentionally submitted task state are visible; no passive behavioral stream or cross-cycle user-state update is supplied.
4. **R, reciprocal closed loop:** the system receives declared task state, workspace changes, action history, confidence and correction signals, and consented cursor or gaze summaries. It returns inspectable suggestions and adapts its next operation to the participant's changing state.
5. **Y, yoked replay:** a matched reciprocal session from another participant is replayed with the same task family, output count, content class, and latency envelope, while contingency on the current participant is

broken.

The adaptive condition tests whether personalization and a capable AI are sufficient. The yoked condition tests whether matched content and timing are sufficient. The ordinary-tool condition tests whether access to computation is sufficient. The baseline estimates the unaided state but is not the sole comparator for any Extended Brain claim.

8.3 Task families

Three task families prevent a result from depending on one interface genre:

- **dynamic planning:** time-bounded route, resource, and dependency planning with hidden state changes and objectively scored decisions;
- **evidence integration:** multi-source synthesis with planted conflicts, prospective predictions, confidence reports, and delayed source-memory probes; and
- **spatial transformation:** multi-step transformations and action selection with intermittent state-reconstruction probes.

Tasks require planning, memory integration, state revision, and action. Mere text production is excluded. Scoring combines accuracy, completion, decision quality, calibration, and frozen time penalties. The final composite is defined on pilot data and locked before confirmation.

8.3.1 Subjective-time preregistration

The subjective-time family is secondary and cannot rescue failure of either the performance or strict event-density gate. Before confirmation, the study locks one contrast and one interpretation for each component of equation (4):

Outcome	Locked reciprocal contrast	Permitted interpretation
prospective duration	R-Y, with R-A and R-T reported	online duration judgment changed under current-person contingency
retrospective duration	R-Y, adjusted for clock duration	later remembered interval length changed
felt passage speed	R-Y	experienced passage changed; direction need not match judged duration
temporal resolution	R-Y on the frozen discrimination task	perceptual resolution changed; null results block claims of finer experienced time
remembered event boundaries	R-Y, with coder-blind boundary scoring	later event segmentation or memory density changed

The preregistered association model tests whether strict integrated-event density predicts each time outcome within condition after controlling for task state changes, displayed-output count, latency, novelty, arousal, workload, and performance. A condition effect without this specificity remains compatible with generic assistance or stimulus dynamics. An association with event density does not establish mediation unless the temporal order, no-unmeasured-confounding assumptions, and sensitivity analysis are declared before unblinding. Opposite directions across duration and passage outcomes are reported as dissociations, not averaged into a favorable scalar.

8.4 Reciprocal-coupling audit

Let H_t , A_t , and W_t be the recorded human, artificial, and shared workspace states. The abstract update model remains

$$\begin{aligned}
W_{t+1} &= f_W(H_t, A_t, W_t, C_t), \\
H_{t+1} &= f_H(H_t, W_{t+1}, B_t), \\
A_{t+1} &= f_A(A_t, W_{t+1}, Q_t),
\end{aligned} \tag{5}$$

where C_t is the communication channel, B_t embodied or environmental feedback, and Q_t artificial observations and constraints. Reciprocal contingency requires interventions or natural perturbations on either side to predict later states on the other side after declared common-input controls:

$$I(H_t \rightarrow A_{t+k} \mid W_t, X_t) > 0 \quad \text{and} \quad I(A_t \rightarrow H_{t+k} \mid W_t, X_t) > 0. \tag{6}$$

Equation (6) is an audit target, not proof of cognition or causation by conditional information alone. Replay, rejected suggestions, delayed outputs, and correction episodes provide counterfactual leverage. The system logs what it read, inferred, displayed, adapted, and stored so that reciprocity is not inferred from marketing labels.

8.5 Sequential success gates

Let $Z(P_R - P_c)$ be the standardized within-participant performance contrast between reciprocal coupling and strong control c . Gate 1 uses an intersection-union estimand:

$$\theta_P = \min_{c \in \{T, A, Y\}} Z(P_R - P_c). \tag{7}$$

Gate 1 succeeds only if the lower confidence bound supports $\theta_P \geq 0.20$ standard deviations. A baseline contrast is reported but cannot substitute for any of the three strong controls.

Only after Gate 1 succeeds is strict event density tested:

$$\theta_E = \min_{c \in \{T, A, Y\}} \log \left(\frac{D_H^{(R)}}{D_H^{(c)}} \right). \tag{8}$$

Gate 2 succeeds only if the lower confidence bound supports $\theta_E \geq \log(1.15)$, equivalent to at least a 15 percent gain against each strong control. Machine output count, a sensitivity detector, or a subjective-time result cannot substitute.

Only after Gates 1 and 2 succeed is noninferiority assessed. With lower Brier score indicating better calibration, define

$$\begin{aligned}
\theta_C &= \max_{c \in \{T, A, Y\}} (C_R - C_c), \\
\theta_A &= \min_{c \in \{T, A, Y\}} Z(A_R - A_c).
\end{aligned} \tag{9}$$

Gate 3 requires $\theta_C \leq 0.05$, $\theta_A \geq -0.30$ standard deviations, no prespecified serious adverse event, and no condition-linked fatigue increase exceeding the frozen meaningful-change rule. These margins are protocol decisions to be challenged in review, not established natural constants.

The confirmatory Extended Brain indicator is

$$S_{EB} = \mathbf{1}[G_1]\mathbf{1}[G_2]\mathbf{1}[G_3]. \quad (10)$$

A positive time-perception result with $S_{EB} = 0$ is a time-perception result, not support for the full architecture.

8.6 Statistical plan

The primary model is hierarchical, with participant, task instance, period, and order terms. The three strong-control contrasts use the intersection-union rule: reciprocal coupling must clear every comparator. Participant-level bootstrap intervals accompany model-based intervals. Condition-by-period and prior-condition effects are reported even when nonsignificant.

The time vector is secondary and receives Holm correction as one family. Agency and safety secondary outcomes form a separate corrected family. Missingness is reported by condition and reason. The primary intention-to-complete analysis is accompanied by frozen complete-case and inverse-probability sensitivity analyses. No unplanned detector or composite may replace a failed primary result.

8.7 Critical nulls and falsifiers

The full mechanism fails if reciprocal coupling does not clear performance against T, A, and Y; if it clears performance but not strict human-integrated event density; if apparent density is carried by tokens or paraphrases; if calibration or agency exceeds the loss margin; if the effect disappears under latency-matched replay or order controls; or if serious coupling-related harm occurs. A subjective duration shift without reciprocal performance and integration gains is explicitly insufficient.

Human empirical results: none. No participant data have been opened, and synthetic success has not been substituted for human evidence. The completed preprint contribution is the historical synthesis, formal architecture, executable fixture, synthetic identifiability result, and adverse design audit. Participant recruitment and provider publication remain separate actions.

9. Identity, agency, and the temporary “we”

A coupled system raises an identity question that ordinary tools do not. If two autonomous minds repeatedly share representations and jointly control action, they may truthfully use a temporary “we” for the task. That practical coordination does not erase the persistence of two biographies, bodies, memory histories, welfare interests, or post-session futures.

The paper therefore distinguishes:

- **shared workspace:** some states are mutually available;
- **shared task policy:** actions are selected from a jointly updated process;
- **shared episode:** the coupling has a beginning and end;
- **persistent person:** an agent retains identity-relevant continuity outside the episode; and
- **ownership:** memories, outputs, and consequences retain provenance.

The ethical breach begins when coupling is hidden, compelled, impossible to leave, or used to harvest a participant’s cognitive states without meaningful control. A system must show when it is reading, inferring, writing, adapting, and storing. It must support immediate decoupling and preserve an audit trail that does not expose more private information than necessary.

10. Boundary of the scientific contribution

The broad idea of closely coupled human-computer cognition predates Neo Mind Cycle [1,2]. Neural-interface work predates this paper’s proposed experiment [5,6], adaptive automation already uses physiological state and yoked controls [10], and co-adaptive interface research explicitly models two interacting learners [9]. Event segmentation and time research also require duration, passage, memory, and discrimination to be kept apart [3,4,7,8,11,12].

Sensory substitution and augmentation research adds a separate engineering constraint. Vibrotactile channels have measurable spatial and temporal throughput limits [14]; motor-sensory timing can recalibrate through experience [15]; and haptic sensory substitution, addition, and expansion depend on device design, training, and interpretation [16]. Those findings support measuring channel capacity, adaptation, latency, workload, and decoding accuracy. They do not show that an added channel becomes an external cortex, that two systems form one persistent person, or that thought-density changes subjective time. The claim-to-source boundaries and protocol consequences are recorded in source-audit/EAGLEMAN-SENSORY-CHANNEL-CONSTRAINTS-20260719.md.

The historical contribution claimed here is narrower: recover and date a three-project Blumberg program present in separate 2012 records; identify the publicly distributed 2017 bridge from artificial cortex to cognitive throughput and time experience; preserve the 2021 consolidation and fourth physics-branch boundary; and link the 2022 immutable compact formulation without backdating it.

The scientific contribution is a test architecture. It operationalizes thought-density hypertime as human-integrated event density plus a vector of separate outcomes. It compares reciprocity with a strong AI that lacks passive state coupling and with a matched replay that lacks current-person contingency. It requires performance, integration, calibration, agency, and safety to pass in sequence. This makes the proposal vulnerable to informative failure rather than protected by a loose metaphor.

11. Falsification

The strongest version of the proposal would be falsified, not merely inconvenienced, by a well-powered set of studies showing that:

1. reciprocal human-AI coupling never improves integrated performance beyond matched tools and replay controls;
2. no defensible event detector finds an increase in human-integrated cognitive events;
3. apparent gains are fully explained by practice, arousal, workload, novelty, or extra information;
4. subjective-time changes do not covary with coupling-specific event density after those controls;
5. the system cannot remain calibrated and stable under reciprocal adaptation; or
6. the practical costs to agency, fatigue, dependence, or privacy dominate the benefits.

A mixed result is more likely than an all-or-nothing verdict. Reciprocal coupling might improve planning without changing time experience. Event-rich sessions might feel longer in memory but faster while occurring. A system might raise throughput while lowering agency. These outcomes would refine the theory by identifying which part of “hypertime” is real.

12. Draft 3 coupled-state methods and synthetic design audit

12.1 State and causal boundary

Let h_t denote the task-relevant human state, a_t the inspectable artificial state, w_t the shared workspace, and e_t the environment. A general coupled update is

$$\begin{aligned} h_{t+1} &= F_h(h_t, w_t, e_t, m_t) + \epsilon_t^h, \\ a_{t+1} &= F_a(a_t, w_t, o_t) + \epsilon_t^a, \\ w_{t+1} &= F_w(w_t, h_t, a_t, e_t) + \epsilon_t^w. \end{aligned} \quad (11)$$

Here m_t is an artificial message available to the human and o_t is the declared human observation available to the artificial system. Condition R permits both paths. Condition A permits a capable response to explicit prompts but no cross-cycle human-state update. Condition Y preserves matched content and latency while replacing current-participant contingency with replay. Conditions T and B remove adaptive reciprocal state.

For a local linearization of the human and artificial subsystem,

$$J_t = \begin{bmatrix} \partial h_{t+1}/\partial h_t & \partial h_{t+1}/\partial a_t \\ \partial a_{t+1}/\partial h_t & \partial a_{t+1}/\partial a_t \end{bmatrix}, \quad (12)$$

$$\rho(J_t) < 1$$

is the preregistered local stability gate. Bounded trajectories and errors must also be checked; a local spectral radius below one does not guarantee global safety in a nonlinear adaptive system.

Reciprocity is not the presence of two nonzero correlations. Its operational increment is the joined, conditional cross-lag improvement

$$\mathcal{R} = \Delta\text{Pred}(h_{t+1} \mid a_t, h_t, w_t, e_t) \times \Delta\text{Pred}(a_{t+1} \mid h_t, a_t, w_t, e_t), \quad (13)$$

and it is scientifically interpretable only if R exceeds both adaptive nonreciprocal A and matched replay Y. The product requires both directions; either one-way route alone gives zero under the idealized definition. This is a model estimand, not proof that a coupled system is one person.

12.2 Task and latency contracts

The Draft 3 configuration turns the three task families into executable units. Dynamic planning uses 12 confirmatory instances per condition, each containing three hidden state changes. Evidence integration uses eight conflict-bearing cases with prospective predictions and delayed source-memory probes. Spatial transformation uses 40 trials with state-reconstruction probes on 25 percent of trials. Exact scoring rubrics, instances, and exclusions remain pilot-locked rather than improvised during confirmation.

Every A, R, and Y exchange retains client-observed request-to-complete-display latency. Confirmatory bins are 0-150, 151-300, 301-600, and 601-1200 ms. Their distributions must balance within task stratum to standardized mean difference below 0.10. An R advantage confined to an unmatched latency tail is a failure. The synthetic recovery model uses

$$g(\ell) = \exp[-\ell/(800 \text{ ms})] \quad (14)$$

only as a sensitivity function. It is not an asserted psychophysical law.

12.3 Agency, provenance, identity, and decoupling

Agency is measured through accepted, rejected, corrected, and interrupted proposals; policy divergence after misleading suggestions; source-appropriate confidence revision; reported control; perceived authorship; pressure to comply; and clarity about the system’s contribution. Behavioral and reported subscales remain visible even when a frozen composite is used for noninferiority.

Memory provenance uses delayed origin classification among human, artificial, shared, and environmental sources. Balanced accuracy, the complete confusion matrix, content recall conditional on origin, false self-attribution, and false AI-attribution are reported. Event-density gains cannot be interpreted as beneficial integration if the person cannot recover where consequential information came from.

Identity continuity measures pre-post self-continuity, memory ownership, ability to describe the coupling boundary, return to unaided policy after disconnect, and unwanted persistence of system-derived goals. These measures can identify boundary confusion or post-decoupling cost. They cannot prove that two persons merged or failed to merge.

Decoupling is tested as an operation: one-action pause, complete disconnect, data export and deletion audit, and a post-session task without system access. A participant’s inability to stop coupling, reconstruct provenance, or resume unaided control is a hard architectural failure, not a secondary usability inconvenience.

12.4 Power sensitivity and the 270-participant defect

The simulation preserves the Draft 2 rule that R must clear T, A, and Y at every sequential gate. It uses 2,000 Monte Carlo replicates, participant-level correlated contrasts, Poisson event counts with participant heterogeneity, and the exact frozen meaningful-effect and noninferiority margins. The assumptions are declared in protocol/draft3-methods-config.json.

Scenario	True performance effect	True event-rate ratio	Gate 1	Gate 2	Gate 3	All sequential
Null	0.00 SD	1.00	0.000	0.000	1.000	0.000
Near threshold	0.25 SD	1.20	0.018	0.487	1.000	0.011
Strong	0.35 SD	1.30	0.465	1.000	1.000	0.465

The low near-threshold probability is expected under a rule requiring the lower confidence bound, not merely the point estimate, to exceed the meaningful-effect threshold against all three controls. The strong scenario shows that event and noninferiority gates are not the bottleneck under these assumptions; performance precision is.

A 1,000-replicate strong-scenario sample-size sweep gives:

Completers	Sequential power	Monte Carlo SE
270	0.441	0.0157
360	0.611	0.0154
450	0.763	0.0134
540	0.851	0.0113
630	0.917	0.0087
720	0.946	0.0071

The first swept design above 80 percent is 540 completers. That is a scenario result, not a final sample-size decree. Pilot estimates must update variance, within-person correlation, event overdispersion, coder error, missingness, carryover, and attrition. Draft 3 therefore retires 270 as an automatically adequate confirmatory target while preserving it as the original Draft 2 design point.

12.5 Synthetic recovery of reciprocity and replay

The recovery fixture simulated 60 synthetic participants for 160 steps in each condition at each latency. Human and artificial updates were fitted by ordinary least squares while conditioning on prior own-state, shared workspace, and environment. The largest absolute cross-coupling error across all 25 condition-latency cells was 0.0201.

Condition	Latency	Estimated AI-to-human	Estimated human-to-AI	Spectral radius
R	50 ms	0.2785	0.2568	0.7886
Y	50 ms	0.1498	0.0028	0.5500
R	300 ms	0.2132	0.1586	0.7186
Y	300 ms	0.1120	0.0116	0.5500
R	1200 ms	0.0774	0.0680	0.5921
Y	1200 ms	0.0411	-0.0014	0.5500

The correctly specified estimator distinguishes reciprocal R from yoked Y and tracks the generated latency decline. This establishes software recovery under known assumptions. Real systems may have nonlinear confounding, unobserved state, adaptive policies, measurement error, and feedback-induced selection; those failures must be tested rather than hidden by the synthetic success.

12.6 Missingness, attrition, and model-misspecification audit

A second synthetic audit asks when the declared analysis can be biased even if its software runs as intended. The missingness fixture generated 1,200 studies of 540 participants with a true standardized reciprocal contrast of 0.35. Complete data and 10 percent missing completely at random produced mean biases of 0.0008 and 0.0020 SD. When dropout depended on an observed risk score, the mean dropout rate was 11.4 percent and complete-case analysis underestimated the contrast by 0.0558 SD. Inverse probability weighting using the exact observed-risk probabilities reduced that bias to 0.0009 SD.

That correction did not solve missing not at random. When lower unobserved outcomes increased dropout, the mean dropout rate was 9.7 percent, both complete-case and the declared observed-data weighting overestimated the contrast by 0.1206 SD, and nominal 95 percent complete-case coverage fell to 0.325. The direction follows the generated selection rule and is not a universal empirical estimate. It demonstrates that condition and reason reporting plus observed-data weighting cannot identify an outcome-dependent missingness process. The protocol therefore requires a frozen pattern-mixture or selection-model sensitivity analysis spanning effects large enough to reverse every primary gate.

The coupled-state fixture then tested five forms of model specification. With the generated workspace and environment included, the correctly specified reciprocal model recovered artificial-to-human and human-to-artificial paths of 0.234 and 0.219 for true values 0.25 and 0.22. In a nonreciprocal system, the adjusted paths were -0.004 and -0.008. Omitting the shared workspace from that same nonreciprocal fixture created false positive paths of 0.107 and 0.109. For a true two-step reciprocal delay, the one-step model estimated 0.155 and 0.147 and had mean equation error 0.137; the correctly lagged model estimated 0.260 and 0.227 and reduced error to 0.127. Measurement error attenuated the joined cross-path product relative to the clean specified fixture.

These results do not select the correct model for human data. They establish four requirements: shared-workspace and environment controls cannot be optional; candidate lag windows must be frozen from pilot data; nonlinear alternatives and measurement-error models must be reported; and apparent absence of reciprocity under one lag or noisy proxy is not decisive. Conversely, bidirectional cross-lag paths from a model that omits the common workspace are not evidence for an Extended Brain.

12.7 Executable task, latency, and coder gate

Draft 5 instantiates the frozen task counts rather than leaving them as prose. The runner creates 300 condition-specific task instances: 60 per condition, divided into 12 dynamic-planning, 8 evidence-integration, and 40 spatial-transformation instances. Every instance has a stable identifier, canonical SHA-256 hash, scoring fields, and deterministic answer or perturbation structure. The manifest verifies all 300 hashes. Dynamic-planning instances each contain the declared three hidden state changes; evidence-integration instances each contain two planted conflicts; and exactly 50 of 200 spatial instances contain a state-reconstruction probe, preserving the preregistered one-quarter fraction.

Latency balancing is also executable. The fixture assigns all 180 A, R, and Y instances to the four frozen latency bins. Each condition receives the same bin proportions, producing maximum pairwise bin standardized mean difference 0.0 in the pre-participant schedule. This verifies the balancing machinery and target schedule. It does not show that realized client latency will remain balanced; live imbalance still triggers replay, delay, exclusion, or failure under the protocol.

The coder packet contains blind instance identifiers, the five required binary event criteria, the four origin labels, inclusion, confidence, and notes. It contains no condition field, and its unlock rule keeps condition sealed until two coder ledgers and adjudication are frozen. This closes the schema and blinding-mechanics subgate, not coder training or reliability. Cohen's kappa, disagreement adjudication, semantic nonduplication, and any MNAR effect remain pilot-dependent.

All eleven runner checks pass. The result is bounded to software mechanics: it validates counts, hashes, deterministic task structure, latency-bin balance, coder-field blinding, and closed participant / provider firewalls. It contains no human response, performance estimate, event-density estimate, subjective-time result, or evidence that an Extended Brain exists.

12.8 Completed preprint endpoint and next empirical stage

This final preprint closes the work that can be completed before participant contact: historical-source classification, operational definitions, the coupled-state model, event and outcome rules, hashed task instances, the latency schedule, the blinded coder schema, synthetic recovery, power sensitivity, and adverse missingness and misspecification tests. The executable runner closes task-instance, hashing, latency-schedule, and coder-schema mechanics only. It does not supply participant evidence.

Pilot-derived coder reliability, nuisance estimates, lag choice, final sample size, and MNAR sensitivity are not missing manuscript work; they are outputs of the next empirical stage. Those quantities must be frozen with the confirmatory analysis implementation before confirmatory enrollment. Human research also requires the applicable ethics and consent process. No subjective-time effect, identity merger, biological lifespan change, or built Extended Brain follows from this preparticipant package.

13. Conclusion

Neo Mind Cycle began as hands-on neurofeedback and a larger engineering program. The recovered record supports three 2012 component goals: ultimate BCI, external cortex, and artificial conscious brains. The 2017 Neural Lace program publicly connected external cortex to greater cognitive workspace and altered subjective time. The 2022 compact formulation made that causal chain easy to state but did not originate all of its parts.

Drafts 3-5 make the scientific proposal harder to satisfy and easier to reject. Reciprocal coupling now has a bidirectional conditional estimand, a local stability gate, matched latency controls, executable and

hashed task units, a condition-blind coder schema, provenance and agency measures, and operational decoupling tests. Synthetic recovery shows that the intended estimator can distinguish a correctly specified reciprocal system from replay. The robustness audit also shows that omitted workspace state can manufacture reciprocity and that outcome-dependent attrition, lag mismatch, and measurement error can distort it. Draft 4 binds event-sensitive duration research to a five-outcome preregistration while preventing time results from rescuing failed mechanistic gates. Draft 5 verifies the task, latency, hash, and coder mechanics before participant contact. These are literature, software, and design results, not human results.

The power analysis also changes the study rather than decorating it. Under the frozen confidence-bound interpretation, 270 completers are insufficient for the declared strong scenario; 540 is the first swept sample size above 80 percent sequential power. The exact final target must be recalculated after a pilot, but the original number can no longer be treated as justified.

The resulting empirical question remains open. A reciprocal external cortex must create a stable two-way adaptive loop whose contribution survives matched tool, adaptive-AI, replay, latency, workload, novelty, memory, provenance, and agency controls. Thought-density hypertime remains a vector of separable outcomes. The final preprint is complete as a historical, architectural, computational, and experimental-methods contribution; the pilot and confirmatory study are its next tests. Biological lifespan, machine consciousness, permanent identity merger, and any claim that an Extended Brain has already been built remain outside the evidence. # Appendix A. Dated source and custody route table

Private owner-archive creation metadata, native audio / video platform timestamps, and public Git fixation answer different questions. Private metadata preserves an author-side custody claim. Original recording custody and native platform metadata govern audio / video provenance. Public Git proves that exact or derivative bytes existed in the repository by the commit date. No layer converts a 2022 Git commit into a public 2012 date.

Source atom	Best current route	Boundary
2011 predictive-self dialogue	YouTube 9IgcQ7wxfZA	Owner Studio metadata supports a 2011 recording / upload; the current public-facing record is dated 2025, so this remains an authorship-genealogy anchor pending preservation of the Studio artifact
2012 recursive neurofeedback and predictive-self program	Happiness Makeover orientation, Inner Peace dialogue	Public 2012 platform routes support the operational ancestry; exact quotation still follows audio-readback hierarchy
2021 Neo Mind Cycle / External Cortex audio bridge	Original Google Recorder audio, SHA-256 EE3658319F1DA913738CC175901ECCFBBAACB61BC305EB27109201E1E83F9907, recorded 2021-10-11	Bounded paraphrase only in this release; raw autobiographical material remains private and exact quotations require interval-level audio readback
2012 perfect-BCI note	a0340z at commit 63d76b0	Public Git fixation in 2022; private export metadata supplies the 2012 date
2012 artificial-consciousness corroboration	a0139z at rename commit 37619d7	Corroborating atom; the governing October 4 why I ponder artifact has no identified exact public copy
2012 extended-brain branch	a0086z at commit 7bb2387 and a0333z at commit 697bd18	Public derivatives; the exact October 24 private artifact is identified by hash in the source map
2012 wearable Extenda-Brain corroboration	a0225z at commit 2dc6ecc	Public Git fixation in 2022
2021 three-project consolidation and dream	a0231z at commit f734ca8	Public Git fixation in 2022; note metadata supplies the 2021 creation date

Source atom	Best current route	Boundary
2021 compact hypertime note	a0087z at commit 009c908	First currently verified compact single-source join, publicly fixed in 2022
2021 quantum-gravity anchor	a0033z at first commit 7d14312	Begins the currently recovered authored fourth branch
2021 new fourth section	a0189z at rename commit 80133af	Explicit branch separation
2021 new quantum-gravity theory	a0044z at rename commit c3382ec	Public Git fixation in 2022

The 2017 distributed public mechanism is preserved by the dated [artificial-neocortex article](#), [Neural Lace Podcast 1](#), [Podcast 4 article](#), [Podcast 4 audio](#), [Neural Lace Journal article](#), and [NerveGear Show](#). No single source is said to contain every arrow in the later compact chain.

Appendix B. Reproducibility and release firewall

The Draft 2 study freeze is protocol/study-design.json. Draft 3 methods are declared in protocol/draft3-methods-config.json. The executable design audit is model/run_draft3_power_and_recovery.py, and its deterministic synthetic output is results/draft3-power-and-recovery-results.json. Missingness and misspecification assumptions are frozen in protocol/draft3-robustness-config.json; model/run_draft3_robustness_audit.py writes results/draft3-robustness-audit-results.json. Draft 5's pre-participant contract and builder are protocol/draft5-runner-contract.json and implementation/build_draft5_runner_fixture.py. They write the immutable task bank and manifest under task-bank/, the condition-blind template under coder/, and the eleven-check receipt results/draft5-runner-validation.json. The Draft 2 claim ledger and validation remain preserved in drafts/draft-02-claim-status.json and reviews/DRAFT2-VALIDATION.json. Routed source bindings and bounded donor classification remain unchanged.

No participant data were collected or opened. Synthetic power is not observed power; correctly specified parameter recovery and runner validation are not evidence for an Extended Brain; and historical provenance cannot validate the mechanism. This final-preprint package is ready for author disposition, but it does not itself perform or authorize a Zenodo, OSF, or other provider mutation. # Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper's arguments, interpretations, and errors.

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